ASP.NET Core Out of Process Hosting Model

Back to ASP.NET Core Web IIS Tutorials Home

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ASP.NET Core, hosting models define how an application runs and interacts with a web server. Broadly, there are two hosting models:

- **In-Process Hosting**: The app runs directly inside the IIS worker process.
- **Out-of-Process Hosting**: The app runs in a separate process, and IIS acts as a reverse proxy.

In this article, we focus on the **Out-of-Process Hosting Model**. In addition, configuration options, troubleshooting, and performance considerations are also discussed. This guide is intended for developers and system administrators who want a clear, practical understanding of ASP.NET Core deployment on Windows.

What is Out-of-Process Hosting?

In Out-of-Process Hosting, your ASP.NET Core application runs as a separate process. The IIS worker process forwards **HTTP requests and responses** to it.

IIS Worker Web Server:

- Receives client requests (URL, SSL, TLS termination, logging, and URL rewriting)
- Acts as the **gateway**, prioritizing your app from client internal requests

Kestrel Listens Only Locally:

- Kestrel binds to localhost (127.0.0.1) and listens on a dynamic port.
- Unauthorized external access attempts are blocked.

This separation of responsibilities ensures robustness. **IIS handles website concerns, while Kestrel handles ASP.NET Core logic.**

Real-World Analogy:

Think of a **restaurant system**:

- **IIS is the front**: Receives customers (requests), manages reservations, and delivers responses.
- **Kestrel is the chef**: Focuses on preparing the food (processing requests).

The kitchen (Kestrel) is hidden, but it works efficiently because the front (IIS) handles all interactions with customers.

Please consider this as introducing a restaurant kitchen or whether to cook food **in-house** or **outsource** it. If you are in **In-Process**, IIS outsources everything to your kitchen. But if you are in **Out-of-Process**, IIS only delivers the raw ingredients (HTTP requests) and your kitchen (Kestrel) does everything else, including cooking the meal. IIS then delivers the meals back to the customers.

Method 1. Editing Project File

In the project's project file (`*.csproj`), add the `<AspNetCoreHostingModel>` element inside your project's `.csproj` file and set its value to `OutOfProcess`. This ensures that the ASP.NET runtime is run in the application outside the IIS worker process, instead in a separate process using the **Kestrel server**.

```
<Project Sdk="Microsoft.NET.Sdk.Web">
  <PropertyGroup>
    <TargetFramework>net8.0</TargetFramework>
    <AspNetCoreHostingModel>OutOfProcess</AspNetCoreHostingModel>
    <Nullable>enable</Nullable>
    <ImplicitUsings>enable</ImplicitUsings>
    <RuntimeIdentifier>win-x64</RuntimeIdentifier>
    <PublishProfile>FolderProfile</PublishProfile>
    <SelfContained>false</SelfContained>
    <ExcludeGeneratedPublishContent>true</ExcludeGeneratedPublishContent>
    <ExcludeGeneratedPublishFiles>true</ExcludeGeneratedPublishFiles>
    <ExcludeGeneratedPublishDirectory>true</ExcludeGeneratedPublishDirectory>
    <ExcludeGeneratedPublishManifest>true</ExcludeGeneratedPublishManifest>
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    <ExcludeGeneratedPublishWebHandlers>true
```